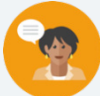


11.2 Key Features of Exponential Functions

Lesson Introduction

 Benchmarks	MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay. MA.912.AR.5.6 Given a table, equation, or written description of an exponential function, graph that function and determine its key features.		 Learning Targets	- I can determine the key features of an exponential function from a table, graph, or exponential function. - I can classify a function as representing exponential growth or decay.
 Benchmark Coherence	Building On N/A		Working Toward MA.912.AR.5.7 Solve and graph mathematical and real-world problems that are modeled with exponential functions. Interpret key features and determine constraints in terms of the context.	
 Essential Questions	- What are the key features of the graph of an exponential function? - How can you determine the key features of an exponential function from a table? - How can you determine the key features of an exponential function from a graph? - How can you determine the key features of an exponential function from an equation?			
 Lesson Narrative	In this lesson, students are introduced to key features of exponential functions. Students will begin by exploring equations, graphs, and tables of exponential functions. They will then progress into determining the key features of exponential graphs from various representations.			
 Component Summary	11.2.1 Warm-Up (~5 min.)	Rhind Papyrus math challenge problem (<i>SE p. 163</i>)	11.2.4 Guided Instruction (10-15 min.)	Formal instruction on determining key features of exponential graphs (<i>SE p. 167</i>)
	11.2.2 Exploration (10 min.)	Explore equations, graph, and tables of exponential functions (<i>SE p. 163</i>)	11.2.5 Your Turn! (5 min.)	Independent practice determining the key features of an exponential function given a graph and a table (<i>SE p. 169</i>)
	11.2.3 Bring It Together (10 min.)	Summarize key features of exponential functions (<i>SE p. 165</i>)	11.2.6 Wrap-Up (~5 min.)	Self-reflection on student understanding of each key feature (<i>SE p. 170</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Asymptote <u>Additional Terms:</u> - Domain - Range - y-intercept - End behavior		N/A	- Students may be confused when identifying intervals where the function is increasing and decreasing when the b value would indicate the opposite (i.e., for reflections). Work through these examples prior to facilitating this lesson to look for these and other areas where misconceptions could occur. (Optional) Word bank for question 5 in 11.2.4 Guided Instruction.

 Teacher Tips	Common Misconceptions - Students may have a difficult time locating the y-intercept from a table when not explicitly given it. - Students may misread a graph in the wrong direction when matching an exponential graph to its function. - Students may struggle with the concept of approaching zero and infinity.	Things to Consider Students are introduced to the concept of an asymptote in this lesson. Since vertical shifts are not in Algebra 1, students will not be exposed to when an asymptote for an exponential isn't $y = 0$. They will further develop this concept in Algebra 2, but the foundation of this concept is laid in this unit.
	Literacy and Language Routines Think-Pair-Share 11.2.3 Bring It Together has students determine which graphs are increasing, decreasing, positive, and negative. Leverage a Think-Pair-Share protocol to allow students the opportunity to try identifying them on their own, and then working with their partners to discuss their thinking and reasonings.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations 11.2.2 Exploration provides students with an opportunity to explore exponential functions represented as graphs, tables of values, and equations. This will allow students to understand the key features of exponential functions represented in multiple ways, including making connections between the different representations.
	 Differentiate Instruction Consider creating an anchor chart that shows the different examples from questions 1-4 from 11.2.4 Guided Instruction. This way students have a reference point for how to determine key features of various exponential functions using different representations as they progress throughout the unit. 11.2.5 Your Turn! provides students with independent practice on determining the key features of an exponential function using a graph and a table. Student performance on this activity can help determine where students may need additional support for future lessons.	
Lesson Notes:		

11.2.1 Warm-Up: Rhind Papyrus

Component Overview: Students will be presented with a math problem presented on the Rhind papyrus.



11.2.1 Warm-Up: Rhind Papyrus

Some of the earliest records of mathematical problem solving come from ancient Egypt in the form of papyri that were written between 1850 and 1600 BCE. One of these, the Rhind Mathematical Papyrus, is a collection of math problems written around 1650 BCE. The problems in the papyrus were used to manage food supplies for cities, among other things.



One of the problems asks: "In a certain village, there were 7 houses; each house had 7 cats; each cat caught 7 mice; each mouse would have (were it not for the cats) eaten 7 ears of spelt; each ear of spelt produced 7 hekats of grain at harvest. How many hekats of grain were saved by the presence of the cats?"

1. Find the solution to the Rhind Papyrus problem.
 $7^5 = 16,807$ hekats of grain were saved.



This problem can be challenging for students. Some students may not recognize this as a multiplication problem and may add instead. Some students will mistake this for the famous riddle "St. Ives," where the trick answer is 1. Consider allowing students to work with a partner to try and solve this problem prior to reviewing. When reviewing the answer, connect this problem to their prior learning with exponents.

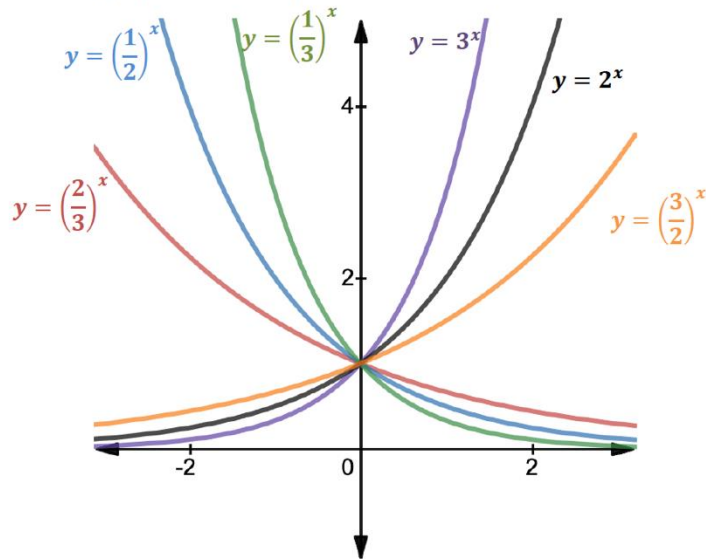
11.2.2 Exploration: Exploring Equations, Graphs, and Tables of Exponential Functions

Component Overview: Students will explore equations, graphs, and tables of exponential functions. Students will review graphs of exponential functions to connect them to what they know about growth and decay.



11.2.2 Exploration: Equations, Graphs, and Tables of Exponential Functions

1. The graphs of six exponential functions are shown.



- a. Work with your partner to complete the table.

	Graphs with a <i>b</i> -value Between 0 and 1	Graphs with a <i>b</i> -value Greater Than 1
Equations	☐ $y = \left(\frac{2}{3}\right)^x$ ☐ $y = \left(\frac{1}{2}\right)^x$ ☐ $y = \left(\frac{1}{3}\right)^x$ ☐ $y = 3^x$ ☐ $y = 2^x$ ☐ $y = \left(\frac{3}{2}\right)^x$	☐ $y = \left(\frac{2}{3}\right)^x$ ☐ $y = \left(\frac{1}{2}\right)^x$ ☐ $y = \left(\frac{1}{3}\right)^x$ ☐ $y = 3^x$ ☐ $y = 2^x$ ☐ $y = \left(\frac{3}{2}\right)^x$
Type of Function	☐ Exponential growth ☐ Exponential decay	☐ Exponential growth ☐ Exponential decay



Before students engage with 11.2.2 Exploration with their partner, consider utilizing a Notice and Wonder routine to offer students the opportunity to observe similarities and differences between the graphs before delving deeper into the activity.



- What do you notice about the shape and structure of the graphs that have a *b*-value between 0 and 1?
- What do you notice about the shape and structure of the graphs that have a *b*-value greater than 1?

11.2.2 Exploration: Exploring Equations, Graphs, and Tables of Exponential Functions (continued)

b. Complete the statements.

For the equations that represent exponential growth, as

the x -values increase, the y -values ☐ increase.
☐ approach zero.

For the equations that represent exponential decay, as

the x -values increase the y -values ☐ increase.
☐ approach zero.



Consider spending a little extra time in discussion on question 1b. Connect the concept of increasing with the x -values to the concept of approaching infinity, as they will use this language in future lessons.

Also, help students to understand what it means to approach zero and why the function will never equal zero.

An **asymptote** is a line that a graph approaches, but never crosses or touches.

c. For the exponential decay equations, as x approaches $+\infty$, what is the horizontal asymptote?
 $y = 0$

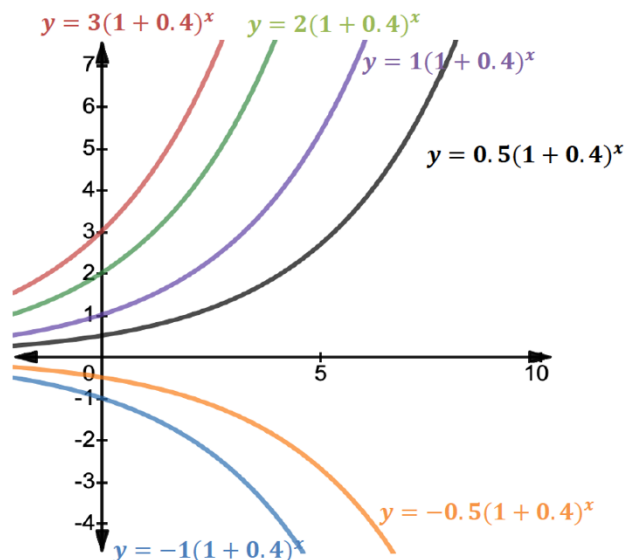
d. For the exponential growth equations, as x approaches $-\infty$, what is the horizontal asymptote?
 $y = 0$

e. What do you notice about the graphs of exponential decay and exponential growth functions as the value of b gets closer to 1?

Answers will vary. I notice when b is a value between 0 and 1, the graph of the exponential decay shows a faster rate of decay the closer the value gets to 1. When b is a value greater than 1, the graph of the exponential growth shows a slower rate of growth the closer the value b gets to 1.

11.2.2 Exploration: Exploring Equations, Graphs, and Tables of Exponential Functions (continued)

2. The graphs of six exponential functions are shown with their corresponding equations.



- What do all the equations have in common?
They are all exponential growth functions with a b value greater than 1.
- What do you notice as the value of a approaches 0?
It gets closer to the x -axis.
- Complete the table to describe the similarities and differences of the graphs of $y = -1(1 + 0.4)^x$ and $y = 1(1 + 0.4)^x$.

Similarities	Differences
<i>Answers will vary. They both have b values greater than 1. They both have a horizontal asymptote of $y = 0$.</i>	<i>Answers will vary. One has the y-values decreasing, and one has the y-values increasing. They both go into different quadrants of the graph.</i>



What do you think the graph of this function would look like? Does it meet the structure of an exponential function? Why might its graph look different?



Consider adding an additional layer for review in question 2. Ask students to determine which of these graphs represent an exponential growth, and which ones represent an exponential decay, including how they can tell.



Elicit responses from various students for the similarities and the differences. Consider creating a t-chart in a centralized location and jot down the information as it is shared so students can follow along.

11.2.3 Bring It Together: Key Features of Exponential Functions

Component Overview: Students will summarize what they have discovered about the key features of exponential functions in the previous component.



11.2.3 Bring it Together: Key Features of Exponential Functions

- Based on the 12 graphs in the Exploration, complete the table.

Key Feature	Question
Domain	Do all of the graphs have a domain of all real numbers? ☐ yes ☐ no
Range	Do all of the graphs have a range of all real numbers? ☐ yes ☐ no
x -intercepts	Do the graphs show an x -intercept? ☐ yes ☐ no
y -intercept	Do the graphs show a y -intercept? ☐ yes ☐ no
Vertex	Do the graphs show a vertex? ☐ yes ☐ no
Axis of Symmetry	Do the graphs show an axis of symmetry? ☐ yes ☐ no



Consider leveraging a polling protocol for this question. Call out a key feature, read the question, and then poll the students in the class to determine if they think yes or no.

11.2.3 Bring It Together: Key Features of Exponential Functions (continued)

Increasing/ Decreasing	Circle the graphs that show intervals where the function is increasing.		
	$y = \left(\frac{2}{3}\right)^x$	$y = \left(\frac{1}{2}\right)^x$	$y = \left(\frac{1}{3}\right)^x$
	$y = (3)^x$	$y = (2)^x$	$y = \left(\frac{3}{2}\right)^x$
	$y = 3(1 + 0.4)^x$	$y = 2(1 + 0.4)^x$	$y = 1(1 + 0.4)^x$
	$y = 0.5(1 + 0.4)^x$	$y = -0.5(1 + 0.4)^x$	$y = -1(1 + 0.4)^x$
	Circle the graphs that show intervals where the function is decreasing.		
	$y = \left(\frac{2}{3}\right)^x$	$y = \left(\frac{1}{2}\right)^x$	$y = \left(\frac{1}{3}\right)^x$
	$y = (3)^x$	$y = (2)^x$	$y = \left(\frac{3}{2}\right)^x$
Positive/ Negative	$y = 3(1 + 0.4)^x$	$y = 2(1 + 0.4)^x$	$y = 1(1 + 0.4)^x$
	$y = 0.5(1 + 0.4)^x$	$y = -0.5(1 + 0.4)^x$	$y = -1(1 + 0.4)^x$
	Circle the graphs that show intervals where the function is positive.		
	$y = \left(\frac{2}{3}\right)^x$	$y = \left(\frac{1}{2}\right)^x$	$y = \left(\frac{1}{3}\right)^x$
	$y = (3)^x$	$y = (2)^x$	$y = \left(\frac{3}{2}\right)^x$
	$y = 3(1 + 0.4)^x$	$y = 2(1 + 0.4)^x$	$y = 1(1 + 0.4)^x$
	$y = 0.5(1 + 0.4)^x$	$y = -0.5(1 + 0.4)^x$	$y = -1(1 + 0.4)^x$
	Circle the graphs that show intervals where the function is negative.		
	$y = \left(\frac{2}{3}\right)^x$	$y = \left(\frac{1}{2}\right)^x$	$y = \left(\frac{1}{3}\right)^x$
	$y = (3)^x$	$y = (2)^x$	$y = \left(\frac{3}{2}\right)^x$
	$y = 3(1 + 0.4)^x$	$y = 2(1 + 0.4)^x$	$y = 1(1 + 0.4)^x$
	$y = 0.5(1 + 0.4)^x$	$y = -0.5(1 + 0.4)^x$	$y = -1(1 + 0.4)^x$



For the increasing/decreasing and positive/negative, have a student provide their answer, and have students agree or disagree with that answer instead of the yes/no option.

11.2.3 Bring It Together: Key Features of Exponential Functions (continued)

End Behavior	What do you notice about the end behavior as x-values get smaller? <i>Answers will vary. For exponential growth, I notice that the graph gets closer to the horizontal asymptote. For exponential decay, I notice that the graph gets further away from the horizontal asymptote.</i> What do you notice about the end behavior as x-values get larger? <i>Answers will vary. For exponential decay, I notice that the graph gets closer to the horizontal asymptote. For exponential growth, I notice that the graph gets further away from the horizontal asymptote.</i>
Asymptotes	What is the horizontal asymptote of exponential growth functions, $f(x) = a(b)^x$ or $f(x) = a(1 + r)^x$? <i>$y = 0$</i> What is the horizontal asymptote of exponential decay functions, $f(x) = a(b)^x$ or $f(x) = a(1 - r)^x$? <i>$y = 0$</i>



For end behavior, have a student provide their answer and have students agree or disagree with that answer instead of the yes/no option.

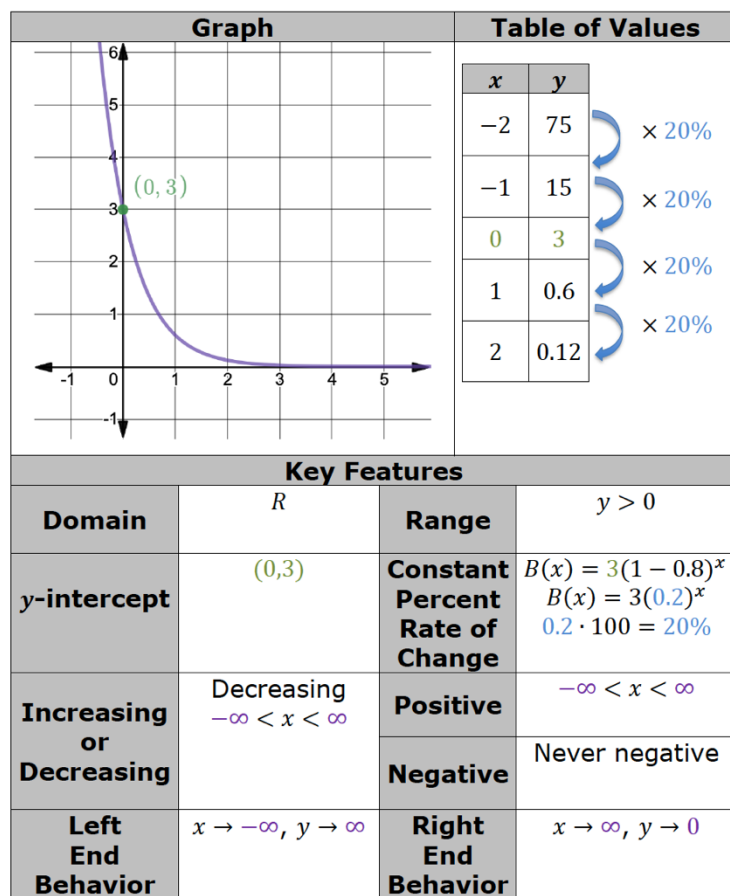
11.2.4 Guided Instruction: Determining Key Features of Exponential Graphs

Component Overview: Students will receive formal instruction on determining key features of exponential graphs given a graph and table of values.



11.2.4 Guided Instruction: Determining Key Features of Exponential Graphs

1. A table and graph of the exponential function $B(x) = 3(1 - 0.8)^x$ are shown. The key features of the graph are color-coded with the key features in the table.



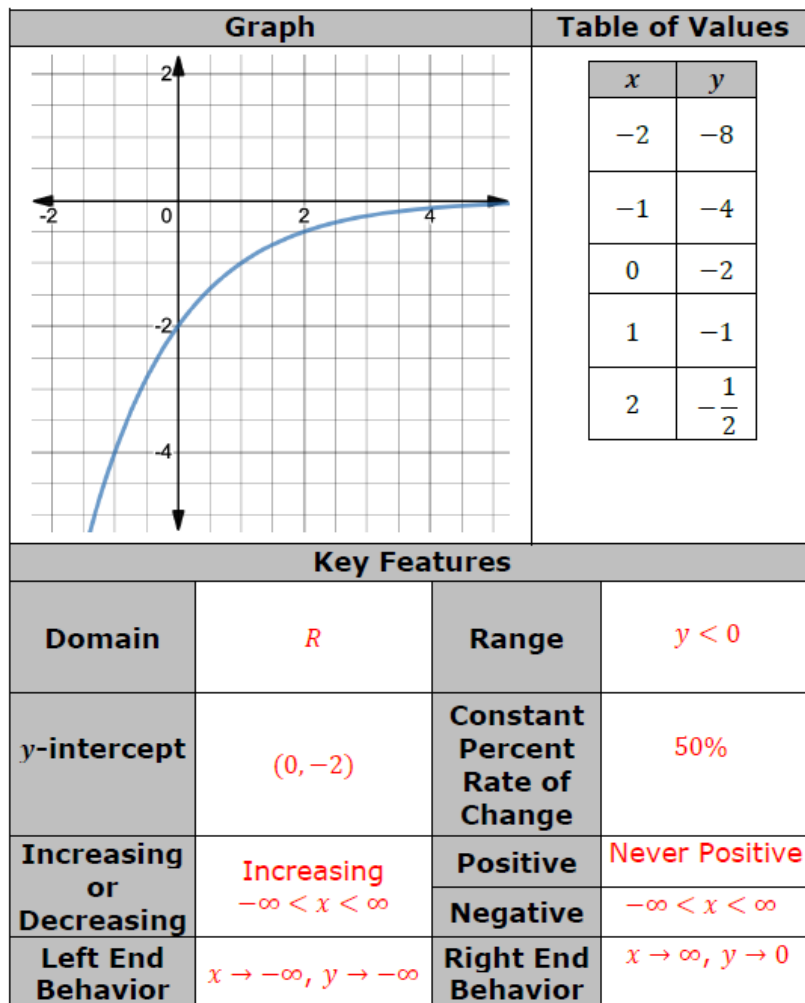
Consider creating an anchor chart for each of these functions to show how to determine the key features for students to be able to refer to throughout the remainder of this lesson and unit.



The purpose of this component is to provide guidance for students in how to determine and express the key features of exponential functions. This should be a guided example with the teacher using Think Aloud and other modeling strategies.

11.2.4 Guided Instruction: Determining Key Features of Exponential Graphs (continued)

- Describe how the constant percent rate of change is found from the table and function.
Using the table of values, divide the y -values to determine the common factor each y -value is multiplied by as the x -value increases by 1. Using the function, identify the value of b in the form $y = a(b)^x$.
- Use the graph and table of $y = -2\left(\frac{1}{2}\right)^x$ to determine the key features.

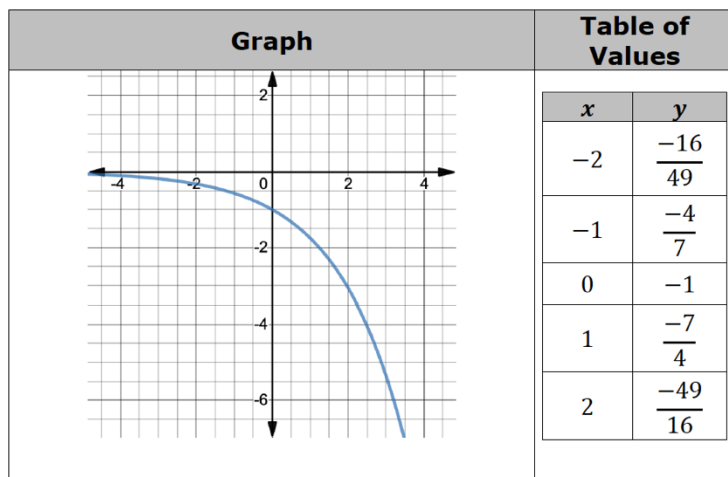


Consider asking the following probing questions before guiding students through this question.

- What do you notice is different about this exponential function compared to the previous one?
- What effect does the -2 have on the graph?
- What effect does the $\frac{1}{2}$ have on the graph?
- How do the key features of this function compare to the previous function?

11.2.4 Guided Instruction: Determining Key Features of Exponential Graphs (continued)

3. Use the graph and table of $y = -(1 + 0.75)^x$ to determine the key features.



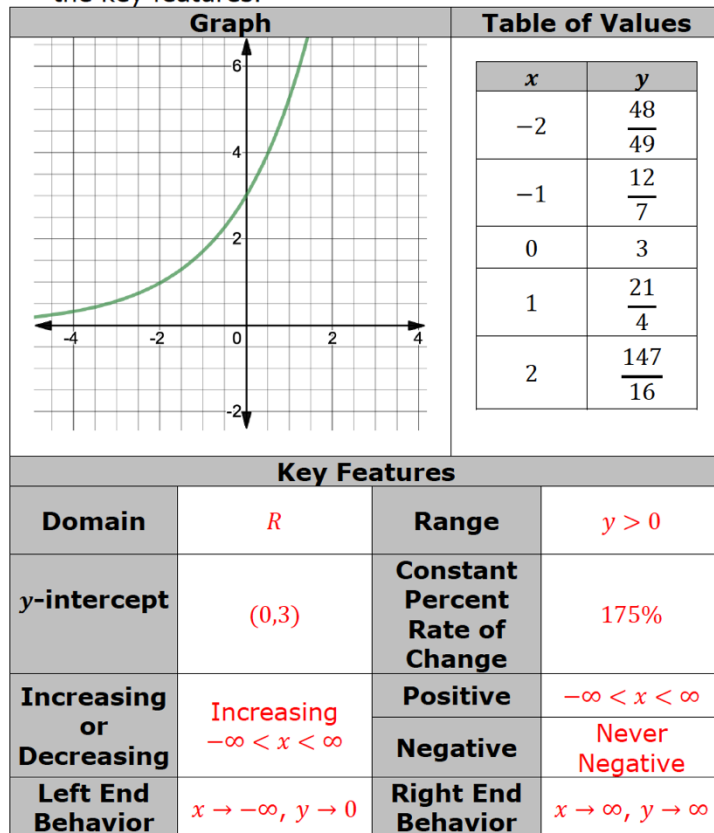
Consider using mark-up strategies to model how to determine the key features from the table of values and from the graph.

- How can the key features be determined from the table of values?
- How can the key features be determined from the graph?
- Which method do you prefer?

Key Features			
Domain	R	Range	$y < 0$
y-intercept	$(0, -1)$	Constant Percent Rate of Change	175%
Increasing or Decreasing	Decreasing $-\infty < x < \infty$	Positive	Never Positive
		Negative	$-\infty < x < \infty$
Left End Behavior	$x \rightarrow -\infty, y \rightarrow 0$	Right End Behavior	$x \rightarrow \infty, y \rightarrow -\infty$

11.2.4 Guided Instruction: Determining Key Features of Exponential Graphs (continued)

4. Use the graph and table of $B(x) = 3\left(\frac{7}{4}\right)^x$ to determine the key features.



Allow informal observation from the first three questions to determine how “guided” this question should be in your classroom. This could mean small-group instruction for some, while others attempt independently or releasing students to try with a partner before going over the responses with the whole group.

11.2.4 Guided Instruction: Determining Key Features of Exponential Graphs (continued)

5. Using the key features found in the previous problems, complete the table.

a is positive or negative	Constant Percent Rate of Change	Increasing or Decreasing	End Behavior as x Increases	End Behavior as x Decreases
Positive	Greater than 100%	Increasing	$x \rightarrow \infty, y \rightarrow \infty$	$x \rightarrow -\infty, y \rightarrow 0$
Positive	Between 0% and 100%	Decreasing	$x \rightarrow \infty, y \rightarrow 0$	$x \rightarrow -\infty, y \rightarrow \infty$
Negative	Greater than 100%	Decreasing	$x \rightarrow \infty, y \rightarrow -\infty$	$x \rightarrow -\infty, y \rightarrow 0$
Negative	Between 0% and 100%	Increasing	$x \rightarrow -\infty, y \rightarrow 0$	$x \rightarrow \infty, y \rightarrow -\infty$



Consider creating this table as an anchor chart for students to reference for the remainder of the unit.

To scaffold this activity, consider providing students with a word bank derived from their work with the previous four graphs and tables.

6. Discuss with your partner what effect the a -value and the constant percent rate of change have on the graph. Write a summary of your discussion.

Answers will vary. The combination of the a -value and the constant rate can determine if the graph will be increasing or decreasing and the end behaviors.

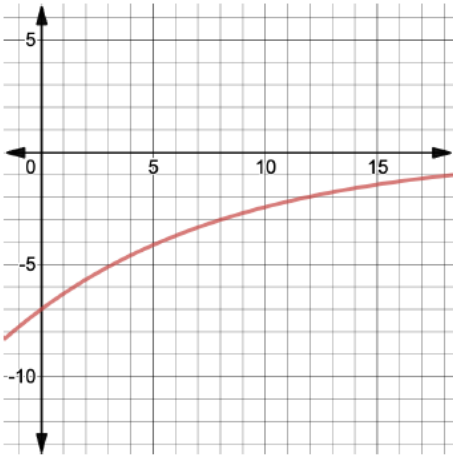
11.2.5 Your Turn!

Component Overview: Students will work independently to determine key features of an exponential function given a graph and a table of values.



11.2.5 Your Turn!

- Use the graph and table of $y = -7(1 - 0.1)^x$ to determine the key features.

Graph		Table of Values													
		<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>$-\frac{700}{81}$</td></tr><tr><td>-1</td><td>$-\frac{70}{9}$</td></tr><tr><td>0</td><td>-7</td></tr><tr><td>1</td><td>$-\frac{63}{10}$</td></tr><tr><td>2</td><td>$-\frac{567}{100}$</td></tr></table>		x	y	-2	$-\frac{700}{81}$	-1	$-\frac{70}{9}$	0	-7	1	$-\frac{63}{10}$	2	$-\frac{567}{100}$
x	y														
-2	$-\frac{700}{81}$														
-1	$-\frac{70}{9}$														
0	-7														
1	$-\frac{63}{10}$														
2	$-\frac{567}{100}$														
Key Features															
Domain	$\mathbb{R}$	Range	$y < 0$												
y-intercept	$(0, -7)$	Constant Percent Rate of Change	90%												
Increasing or Decreasing	Increasing $-\infty < x < \infty$	Positive	Never Positive												
		Negative	$-\infty < x < \infty$												
Left End Behavior	$x \rightarrow -\infty, y \rightarrow -\infty$	Right End Behavior	$x \rightarrow \infty, y \rightarrow 0$												



Consider leveraging a Think-Pair-Share protocol with this activity. Allow students several minutes to try this on their own and then work with a partner to compare their answers and process with one another to come to a consensus on the final answer.

11.2.6 Wrap-Up: Reflection

Component Overview: Students will complete a reflection where they rate their levels of understanding of each key feature from this lesson. This activity should be done independently to review student confidence from each student.



11.2.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each key feature.

Answers will vary.



I need help.



I got it.



I could help someone.

Finding domain and range of an exponential function			
Finding the constant percent rate of change			
Describing the end behavior of an exponential function			
Finding the y-intercept			
Finding the intervals where the function is positive or negative			
Finding the intervals where the function is increasing or decreasing			
Identifying the horizontal asymptote			



Consider pairing students to discuss their reflections to see if peers can answer any questions they may have using examples from the lesson.